

Project Title	Multi-Brand Marketing Campaign Performance Analysis and Prediction Using Python, SQL, and Machine Learning
Skills take away From This Project	• Python & Data Processing • CSV to DataFrame Conversion • Data Cleaning (Handling Missing Values & Duplicates) • Data Preprocessing & Transformation • Multi-Label Encoding (Channel_Used Feature Engineering) • Feature Engineering (ROI Validation & Profit Flag Creation) • Exploratory Data Analysis (EDA) • Marketing Performance Analysis (Campaign, Channel & Brand-wise Insights) • SQL & MySQL Queries • Regression Modeling (Revenue / ROI Prediction) • Classification Modeling (Profit vs Loss Prediction) • Model Evaluation (Accuracy, RMSE, Precision, Recall) • Data Visualization (Seaborn / Matplotlib / Plotly) • Dashboard Development (Streamlit) • End-to-End Machine Learning Pipeline
Domain	Marketing Analytics & Business Intelligence

Problem Statement:

Marketing teams across multiple brands such as Nykaa, Purplle, and Tira generate large volumes of campaign data, including impressions, clicks, conversion rates, acquisition costs, revenue, and ROI. However, this data is often stored in raw and complex formats like CSV files, making it difficult to analyze and extract meaningful insights.

Additionally, challenges such as missing values, inconsistent ROI calculations, and multiple marketing channels within a single column increase the complexity of data preprocessing and impact the reliability of analysis.

The objective is to transform this raw data into a structured format through data cleaning, preprocessing, and exploratory data analysis (EDA). This includes handling

missing values, applying multi-label encoding for categorical features, and creating new features such as profit or loss indicators based on ROI.

This project aims to build a complete **end-to-end machine learning pipeline** that enables candidates to:

- Clean and preprocess marketing campaign data
- Handle missing values and perform data transformation
- Apply multi-label encoding for features like marketing channels
- Perform exploratory data analysis (EDA)
- Engineer new features such as Profit/Loss classification
- Build regression models to predict revenue
- Develop classification models to predict campaign success (profit vs loss)
- Evaluate model performance using appropriate metrics

Objective:

The objective of this project is to design and implement a Marketing Campaign Performance Prediction System that follows the complete machine learning lifecycle:

1. Data Collection
2. Data Cleaning & Preprocessing
3. Exploratory Data Analysis (EDA)
4. Feature Engineering (Multi-label Encoding, Profit/Loss Creation)
5. Model Building (Regression & Classification)
6. Model Evaluation and Performance Analysis

Approach:

1. Data Collection

- Import the dataset in CSV format
 - Load the dataset into Python using Pandas DataFrame
 - Understand the structure, columns, and data types
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2. Data Preprocessing

- Handle missing values (null data)
 - Remove duplicate records
 - Perform data type conversion
 - Clean and standardize the dataset
 - Validate and correct ROI values (if required)
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3. Feature Engineering

- Create new feature: Profit/Loss flag based on ROI
 - Apply multi-label encoding for Channel_Used column
 - Select relevant features for modeling
 - Perform feature transformation if needed
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4. Exploratory Data Analysis (EDA)

- Analyze campaign performance across brands
 - Identify top-performing and low-performing campaigns
 - Explore relationships between spend, clicks, revenue, and ROI
 - Analyze channel-wise effectiveness
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5. Model Building

- Split the dataset into training and testing sets to evaluate model performance
- Build regression models to predict Revenue based on campaign features
- Build classification models to predict Profit/Loss using the engineered Profit_Flag variable
- Train models using appropriate algorithms such as Linear Regression, Logistic Regression, Decision Tree, and Random Forest
- Perform feature selection and ensure no data leakage (exclude ROI from classification features)
- Tune model parameters to improve performance
- Validate models using unseen test data

6. Model Evaluation

- Evaluate regression models using RMSE, MAE, R^2 , MSE
- Evaluate classification models using Accuracy, Precision, Recall, F1-score
- Compare model performance and select the best model

7. Insights & Reporting

- Generate insights on campaign performance
- Identify key factors affecting profitability
- Provide data-driven recommendations for marketing strategies
- Support decision-making using model predictions

8. Application Development

Build an interactive user interface using Streamlit to:

1. Input campaign details (e.g., impressions, clicks, spend, channels, etc.)
2. Apply the same preprocessing and feature engineering steps (encoding, transformations)
3. Display predicted Revenue (Regression Output)
4. Display predicted Profit/Loss (Classification Output)
5. Visualize key inputs and prediction results for better understanding

Dataset Description:

The dataset contains detailed information about marketing campaigns, customer engagement, and financial performance:

Campaign_ID: Unique identifier for each campaign.

Campaign_Type: Type of marketing campaign (e.g., awareness, conversion, promotion).

Target_Audience: Group of customers targeted by the campaign.

Duration: Total duration of the campaign (in days).

Channel_Used: Marketing channels used for the campaign (may contain multiple channels).

Impressions: Number of times the campaign was displayed to users.

Clicks: Number of users who clicked on the campaign.

Leads: Number of potential customers generated.

Conversions: Number of successful actions (e.g., purchases or sign-ups).

Revenue: Total revenue generated from the campaign.

Acquisition_Cost: Cost incurred to run the campaign.

ROI: Return on Investment indicating campaign profitability.

Language: Language used in the campaign content.

Engagement_Score: Measure of user interaction and engagement level.

Customer_Segment: Category of customers targeted (e.g., new, returning, premium).

Date: Date on which the campaign was executed.

Dataset Link: [Marketing Campaign Performance Prediction Datasets](#)

Results:

- Clean and preprocessed marketing dataset (handled missing values & duplicates)
- Successful CSV to DataFrame conversion using Python
- Applied multi-label encoding for marketing channels
- Engineered new feature (Profit/Loss classification based on ROI)
- Performed effective exploratory data analysis (EDA) on campaign performance
- Built regression models to predict Revenue and ROI
- Developed classification models to predict campaign Profit/Loss
- Achieved reliable model performance using evaluation metrics
- Deployed machine learning models using Streamlit application
- End-to-end machine learning pipeline implementation

Project Evaluation metrics:

- Proper data preprocessing (handling missing values & duplicates)
- Accurate CSV to DataFrame conversion
- Effective Exploratory Data Analysis (EDA)
- Correct feature engineering (multi-label encoding, Profit/Loss creation)
- Proper handling of categorical and numerical features
- Effective model building (Regression & Classification)
- Model performance evaluation using appropriate metrics (Accuracy, RMSE, Precision, Recall, F1-score)
- Avoidance of data leakage (e.g., excluding ROI in classification features)
- Efficient model selection and tuning
- Successful deployment using Streamlit application
- Clean and well-structured Python code
- Insight generation and interpretation based on model outputs

Technical Tags:

Python, CSV Data Handling, Data Preprocessing, Exploratory Data Analysis (EDA), Feature Engineering, Multi-Label Encoding, Machine Learning, Regression Models, Classification Models, Model Evaluation, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit, Marketing Analytics, Campaign Performance Analysis, Business Insights.

Project Deliverables:

- CSV dataset (raw data file)
- Cleaned and preprocessed dataset (CSV / DataFrame)
- Feature engineered dataset (including encoded features and Profit/Loss column)
- Python script / Jupyter Notebook for data preprocessing and model building
- Regression model for Revenue / ROI prediction
- Classification model for Profit/Loss prediction
- Model evaluation report (Accuracy, RMSE, Precision, Recall, F1-score)
- Streamlit application for model deployment and predictions
- EDA report with insights
- Final project documentation file

Project Guidelines:

- Follow proper data preprocessing (handle missing values & duplicates)
- Ensure correct CSV to DataFrame conversion
- Perform meaningful Exploratory Data Analysis (EDA)
- Apply proper feature engineering (multi-label encoding, Profit/Loss creation)
- Handle categorical and numerical features appropriately
- Avoid data leakage during model building (exclude ROI in classification)
- Use appropriate machine learning algorithms for regression and classification
- Evaluate models using suitable metrics (Accuracy, RMSE, Precision, Recall, F1-score)
- Maintain clean, readable, and modular Python code
- Build a user-friendly Streamlit application for predictions
- Generate clear insights from model outputs
- Provide well-structured documentation

References:

Project Live Evaluation Guidelines	<u>Project Live Evaluation</u>
Project Excellence Series: Guided Learning & Problem Solving [Python & SQL](Tamil)	<u>Project Excellence Series: Guided Learning & Problem Solving [Python & SQL](Tamil)</u>
Capstone Explanation Guideline	<u>Capstone Explanation Guideline</u>
GitHub Reference	<u>How to Use GitHub.pptx</u>
Streamlit Reference English	<u>Streamlit Recordings</u>

Streamlit Reference Tamil	<u>Streamlit Recording Tamil</u>
Project Orientation (English)	
Project Orientation (Tamil)	

Timeline:

The project must be completed and submitted **within 10 days from the assigned date.**

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

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About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

Note: You can join the doubt session using the link provided in sheet given below

Timing: Monday-Saturday (3:30PM to 4:30PM)

Session Link : [Project Doubt Session \[DS/AIML\]](#)

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing: Monday-Saturday (05:30PM to 07:00PM)

Booking link : <https://forms.gle/1m2Gsro41fLtZurRA>

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